

BIS Gold Jewellery Purity & Consumer Compensation Reference

Structured knowledge source for BIS MITRA AI-agent demonstration

DEMO KNOWLEDGE REFERENCE

Based on BIS source material. This document combines verified BIS information with clearly isolated Clone B demonstration data.

Important: ₹7,000/g is a Clone B demonstration input, not an official BIS gold rate. The transaction example and its testing charge are demo values.

1. DOCUMENT METADATA

Field	Value
Title	Gold Jewellery Purity & Consumer Compensation Reference
Domain	BIS Hallmarking / Gold Jewellery / Consumer Protection
Purpose	Reference material for BIS MITRA demonstration
Source Authority	Bureau of Indian Standards (BIS)
Relevant Reference	BIS Rules, 2018 — Rule 49; IS 1417:2016 — Gold and Gold Alloys, Jewellery/Artefacts — Fineness and Marking — Specification
Document Status	Demo Knowledge Reference — based on BIS source material
Document ID	BIS-GOLD-COMP-001
Version	v1.0
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Scope note. The document is designed for retrieval-augmented generation (RAG). Each major rule or factual statement includes a source/reference field, while demo-only values are explicitly labeled.

2. GOLD PURITY AND FINENESS REFERENCE

Carat describes the gold grade used for jewellery. Fineness expresses gold content in parts per thousand (ppt). For example, 916 means 916 parts gold per 1,000 parts of the alloy; 750 means 750 parts gold per 1,000 parts of the alloy.

Carat	Fineness	Purity
14K	585	58.5%
18K	750	75.0%
20K	833	83.3%
22K	916	91.6%
23K	958	95.8%
24KS	995	99.5%

Extraction rule: Fineness is the numerical gold-content value in parts per thousand. Purity percentage = fineness ÷ 10.

Source / Reference: BIS Hallmarking FAQs; IS 1417:2016. BIS FAQ states that IS 1417:2016 permits the six caratage/fineness grades shown above. See Authoritative References.

2.1 Terminology for RAG

Term	Definition	Example
Carat	Gold grade expressed on the carat scale.	22K
Fineness	Gold content in parts per thousand.	916
Purity	Gold content expressed as a percentage.	91.6%

Source / Reference: BIS Hallmarking FAQs; IS 1417:2016.

3. GOLD HALLMARK COMPONENTS

Since the introduction of HUID for gold jewellery, BIS describes the hallmark as three marks: BIS logo, purity in caratage/fineness, and a six-digit alphanumeric HUID number.

Illustrative hallmark format

BIS Logo
22K916
A1B2C3

22K = caratage; 916 = fineness; A1B2C3 = HUID.

Component	Meaning	RAG interpretation
BIS Standard Mark / Logo	BIS mark forming part of the hallmark.	Indicates the BIS hallmark mark component.
Purity / Fineness Mark	Caratage together with fineness.	Example: 22K916.
Six-digit alphanumeric HUID	Hallmark Unique Identification number; unique for each hallmarked item and traceable.	Example: A1B2C3. Can be checked using the BIS CARE App’s Verify HUID feature.

Source / Reference: BIS Hallmarking FAQs — HUID definition and three-mark hallmark; BIS Consumer Protection — Verify HUID.

3.1 HUID verification

BIS states that consumers can use the BIS CARE App and its “Verify HUID” feature to verify authenticity/details associated with a hallmarked gold jewellery item. The BIS FAQ also states that HUID can provide information related to the jeweller’s registration number.

Source / Reference: BIS Hallmarking FAQs; BIS Consumer Protection information.

3.2 Important distinction for retrieval

The example HUID A1B2C3 above is an illustrative value, not a real BIS record. The document does not claim that this HUID exists in BIS systems.

4. CONSUMER GOLD TESTING

BIS consumer-protection information states that a consumer can get jewellery/sample tested at a BIS-recognized Assaying & Hallmarking Centre (A&H; Centre). The centre undertakes testing on a chargeable basis and issues an Assay Report with identification details of the article.

4.1 Consumer testing workflow

Step	Action	Evidence / Output
1	Consumer suspects that actual purity is lower than the declared/marked purity.	Hallmarked article + relevant purchase/identification information.
2	Take the jewellery/sample to a BIS-recognized A&H; Centre for testing.	Testing request at the recognized centre.
3	The article/sample is tested using the applicable procedure supported by the centre/BIS guidance.	Measured purity / assay result.
4	An Assay Report is issued with identification details as applicable.	Assay Report / test result.
5	If the hallmarked article is found to have lower purity than marked, the tested purity becomes an input to the applicable compensation assessment.	Declared fineness, tested fineness, weight and testing-charge information.

4.2 Testing methods explicitly supported by BIS material

BIS consumer-testing guidance for old/unhallmarked gold states that each jewellery piece should be individually checked by XRF for gold purity and prohibited elements, and that sampling should minimize damage with consumer consent. BIS FAQ material also references fire assay under IS 1418 for consumer testing at recognized A&H; Centres.

Source / Reference: BIS Guidelines on Testing of old gold lying with consumers from BIS recognized AHC; BIS Hallmarking FAQs; IS 1418:2009.

Do not hard-code a universal testing fee. BIS web material can contain location/method-specific testing charges. In this demo document, the ₹1,800 testing-charge value is a Clone B transaction input and must not be treated as a universal BIS fee.

5. RULE 49 — COMPENSATION FOR PURITY SHORTFALL

Rule 49 principle

BIS states that, for a precious-metal article that does not conform to the relevant standard, compensation to the buyer/customer is two times the amount of the difference calculated on the basis of shortage of purity for the weight of the article sold, together with the testing charges.

Source / Reference: BIS Rules, 2018 — Rule 49(1), as reproduced/confirmed in BIS Consumer FAQ and BIS A&H; Centre FAQ.

5.1 Calculation representation for this demo

The following mathematical representation is a demo calculation model translating the Rule 49 principle into explicit RAG/agent steps. It is not a substitute for the legal text.

Purity Shortage = Declared Fineness – Tested Fineness

Purity Shortage Fraction = (Declared Fineness – Tested Fineness) / 1000

Purity Deficit Value = Weight × Purity Shortage Fraction × Applicable Gold Rate

Compensation = 2 × (Purity Deficit Value + Testing Charges)

Legal wording versus calculation model: Rule 49 itself describes compensation as two times the amount of difference calculated on the basis of shortage of purity for the weight sold and the testing charges. The explicit “gold rate” variable in the formula above is an implementation input for the demo calculation and must be supplied by a verified transaction/data source; it is not presented here as a BIS-set gold rate.

Source / Reference: BIS Consumer FAQ; BIS A&H; Centre FAQ; BIS Rules, 2018 — Rule 49(1).

6. REQUIRED INPUTS FOR THE CALCULATION ENGINE

The calculation engine should treat each field as an explicit input. It must not invent missing values.

Input	Meaning	Example
Weight	Weight of jewellery sold.	20 g
Declared Fineness	Fineness marked/declared on the article.	916
Tested Fineness	Fineness determined by testing.	750
Gold Rate	Applicable rate supplied for the calculation.	₹7,000/g
Testing Charges	Applicable testing charge used for this transaction.	₹1,800

DEMO / CLONE B — RATE AND TRANSACTION INPUTS

Gold Rate Source Type: DEMO / CLONE B

₹7,000/g is a demonstration transaction input and is NOT an official BIS gold rate.

Testing Charges ₹1,800 in the worked case are also treated as transaction-level demo data. Do not generalize this value to all BIS-recognized testing facilities or all testing methods.

6.1 Input validation rules

Condition	Agent action
Declared fineness missing	Ask for the marked/declared fineness; do not assume 916, 750, or another grade.
Tested fineness missing	Ask for the test/assay result; do not infer purity from appearance or carat alone.
Weight missing	Ask for the jewellery weight before calculating compensation.
Gold rate missing	Ask for the applicable rate for the relevant transaction/calculation date and identify its source.
Testing charges missing	Ask for the applicable transaction testing charge or mark the calculation incomplete.

7. DEMO TRANSACTION RECORD

DEMO / CLONE B — GOLD-COMP-001

Product: Gold Jewellery

Weight: 20 grams

Declared Purity: 22K / 916

Tested Purity: 18K / 750

Applicable Gold Rate: ₹7,000 per gram

Testing Charges: ₹1,800

Data Source: Clone B Demo Transaction Dataset

7.1 Data provenance

Field	Value	Authority / source type
Purity grades	14K585, 18K750, 20K833, 22K916, 23K958, 24KS995	BIS / IS 1417:2016
HUID structure	Six-digit alphanumeric HUID; part of three-mark hallmark	BIS Hallmarking FAQs
Consumer testing	Recognized A&H; Centre; Assay Report	BIS Consumer Protection
Rule 49 principle	Two-times shortfall difference plus testing charges	BIS Rules, 2018 / BIS FAQ
₹7,000/g gold rate	Demo input	Clone B — not BIS
₹1,800 testing charge	Demo transaction input	Clone B — not a universal BIS fee

7.2 Interpretation constraint

This transaction demonstrates how a RAG/agent system can combine authoritative BIS rules with external transaction inputs. It does not establish that ₹7,000/g is a BIS rate, nor that ₹1,800 is the universally applicable testing charge.

8. COMPLETE WORKED CALCULATION

DEMO CALCULATION — Gold rate and transaction values supplied by Clone B

Case ID: GOLD-COMP-001 | Weight: 20 g | Declared: 916 | Tested: 750

Gold rate: ₹7,000/g | Testing charges: ₹1,800

STEP 1 — Purity Shortage

$$916 - 750 = 166$$
$$\text{Purity shortage} = 166 / 1000 = 0.166 = 16.6\%$$

STEP 2 — Purity Deficit Value

$$20 \times 0.166 \times ₹7,000 = ₹23,240$$

STEP 3 — Add Testing Charges

$$₹23,240 + ₹1,800 = ₹25,040$$

STEP 4 — Apply 2× Compensation Factor

$$₹25,040 \times 2 = ₹50,080$$

Demo output	Value
Estimated Compensation	₹50,080
Purity Deficit Value	₹23,240
Testing Charges	₹1,800
Compensation Multiplier	2×

Final demo result: ₹50,080. This result follows the user-specified demo formula and demo inputs. It is not a statement of an official BIS gold rate or a universal BIS testing fee.

Source / Reference: Rule 49 compensation principle: BIS Rules, 2018 / BIS Consumer FAQ. Numerical inputs: Clone B Demo Transaction Dataset.

9. BIS MITRA AGENT CALCULATION WORKFLOW

Use the following ordered workflow when answering a compensation-calculation query.

Step	Agent instruction
1	Identify declared fineness.
2	Identify tested fineness.
3	Identify jewellery weight.
4	Identify applicable gold rate.
5	Identify testing charges.
6	Calculate purity shortage.
7	Calculate purity deficit value.
8	Add testing charges.
9	Apply the applicable Rule 49 compensation calculation.
10	Show intermediate calculations.
11	Cite the relevant BIS source.
12	Clearly state the source of the gold-rate input.

9.1 Missing-input behavior

User situation	Agent response pattern
Weight missing	"I need the jewellery weight to calculate the compensation."
Gold rate missing	"I need the applicable gold-rate input for the relevant date. This value should come from Clone B or another verified source."
Other required input missing	Ask specifically for the missing value; do not invent or substitute a default.

9.2 Source-selection behavior

When the answer is about a BIS rule, purity grade, HUID, or consumer-testing process, retrieve the authoritative BIS source. When the answer requires a transaction value such as a gold rate or demo testing charge, retrieve the transaction/dataset source and explicitly identify it as such.

RAG safety rule: Never convert a Clone B value into an official BIS fact merely because it appears next to a BIS rule.

10. SUPPORTED USER QUERIES

The following examples are intended as retrieval and agent-behavior test cases.

Category	Example user query	Expected retrieval / behavior
Knowledge	"What does 22K916 mean?"	Explain caratage 22K and fineness 916; cite BIS/IS 1417.
Knowledge	"What is the difference between 22K916 and 18K750?"	Compare fineness and purity percentage without inventing additional rules.
HUID	"How does HUID verification work?"	Explain six-digit alphanumeric HUID and BIS CARE Verify HUID feature; cite BIS.
Testing	"Where can I get my gold jewellery tested?"	Explain testing at a BIS-recognized A&H; Centre and the Assay Report.
Compensation	"My jewellery is marked 916 but tested at 750. Can I claim compensation?"	Explain the Rule 49 principle and required inputs; avoid promising a specific amount without complete data.
Calculation	"I have 20g jewellery marked 916 and tested at 750. Calculate the compensation."	Request gold rate and testing charges if absent; use the demo case only when clearly identified as the demo.
Missing Input	"Calculate my compensation."	Ask for weight, declared fineness, tested fineness, applicable gold rate, and testing charges.

10.1 Retrieval chunking guidance

Recommended semantic chunks: purity/fineness table; hallmark/HUID components; consumer-testing workflow; Rule 49 principle; calculation formula; input schema; demo transaction; worked calculation; agent workflow; source registry. Preserve each source/reference line with its associated chunk.

This structure reduces the risk that a model retrieves the ₹7,000/g demo rate without also retrieving its "DEMO / CLONE B" provenance label.

11. AUTHORITATIVE REFERENCES

The following references are official BIS or Government of India sources used to construct this document. URLs are included for provenance and downstream retrieval.

ID	Document / Page	Relevant use	URL
S1	BIS Hallmarking FAQs — General	HUID; three-mark hallmark; permitted gold grades; consumer testing; hallmarking information	https://www.bis.gov.in/hallmarking-overview/hallmarking-faqs/hallmarking-faq/?lang=en
S2	BIS Hallmarking FAQs — Consumers	Rule 49 compensation principle; complaint information; HUID verification	https://www.bis.gov.in/hallmarking-overview/hallmarking-faqs/mandatory/?lang=en
S3	BIS Consumer Protection — Hallmarking	Consumer testing at BIS-recognized A&H; Centres; Assay Report; compensation; Verify HUID	https://www.bis.gov.in/hallmarking-overview/consumer-protection/?lang=en
S4	BIS A & H Centre FAQ	Rule 49 text in BIS FAQ context; complaint redressal; recognized A&H; Centre information	https://www.bis.gov.in/a-h-centre/?lang=en
S5	BIS Act, Rules & Regulations	Official BIS repository; BIS Rules, 2018 incorporating amendments	https://www.bis.gov.in/the-bureau/bis-act-rules-and-regulations/?lang=en
S6	BIS Summary of Indian Standards — IS 1417:2016	IS 1417 title, gold fineness grades, related assay/testing standards, HUID context	https://www.services.bis.gov.in/tmp/tbl5_2024-11-10_1124.pdf
S7	BIS Guidelines on Testing of old gold lying with consumers	Consumer testing workflow; XRF checking; sampling guidance	https://www.bis.gov.in/wp-content/uploads/2022/01/Guidelines-on-testing-of-unhallmarked-gold-jewellery-and-artefacts-of-the-consumer-by-the-Assaying-and-Hallmarking-Centres.pdf
S8	BIS LIMS — IS 1417 testing facilities	Official BIS laboratory scope and testing-charge listings; demonstrates that charges can be facility/context-specific	https://lims.bis.gov.in/home/search_is_number/?is_number__doc_no=1417

Note on IS 1417 access: BIS's public summary identifies IS 1417:2016 as "Gold and gold alloys, jewellery/ artefacts — Fineness and marking — Specification" and lists the relevant gold fineness grades. The full Indian Standard may be subject to BIS standards-access controls.

Note on Rule 49: The current BIS web pages used for this document reproduce/confirm the Rule 49 compensation principle. The official BIS "Act, Rules & Regulations" page is the repository for the Rules.

12. RAG METADATA

Structured metadata for indexing, retrieval, entity extraction, and calculation-agent routing.

document_id: BIS-GOLD-COMP-001 domain: gold_jewellery subdomain: purity_compensation entities: gold hallmark HUID fineness purity Rule 49 assaying testing compensation calculation_inputs: weight declared_fineness tested_fineness gold_rate testing_charges calculation_outputs: purity_shortage purity_deficit_value compensation data_sources: BIS official sources Clone B demo transaction data demo_only_fields: gold_rate transaction_id testing_charges_if_not_officially_sourced

12.1 Final quality controls

Check	Status
1	Compensation formula is internally consistent with the demo calculation representation.
2	Demo calculation: ₹50,080.
3	₹7,000/g is explicitly identified as Clone B demo data and not an official BIS gold rate.
4	Official BIS information and demo data are separated.
5	Rule-based claims have BIS source/reference fields.
6	The document is structured into retrieval-friendly sections and explicit input/output fields.
7	No universal BIS testing fee is asserted; the ₹1,800 value is treated as demo transaction data.

DEMO DATA BOUNDARY

The worked calculation is a demonstration of the agent workflow. ₹7,000/g, the transaction identifier, and the ₹1,800 testing-charge value are not to be promoted into general BIS facts.

For real-world use, the agent should retrieve the applicable transaction inputs from a verified source and cite the source separately from the BIS rule.